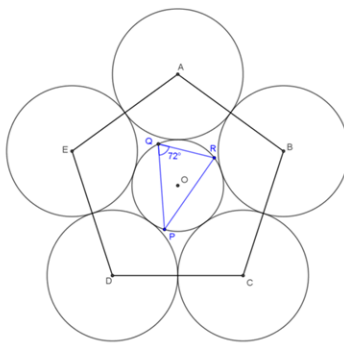


Question 18 – Stage 1



Circles with unit radius are centred on the vertices of a regular pentagon, $ABCDE$, of side length 2 units.

A sixth circle, centre O , touches each of these circles, as shown.

Triangle PQR has its vertices on this circle.

If angle PQR is 72° , what is the length (to 3d.p.) of side PR ?

Working

$$\begin{aligned}
 \angle DOC &= \frac{360}{5} = 72 \\
 \angle ODC = \angle DCO &= \frac{180 - 72}{2} = 54 \\
 DO &= \frac{1}{\cos(54)} \\
 OP = OR &= \frac{1}{\cos(54)} - 1 \\
 \angle POR &= 144 \\
 PR &= \sqrt{OP^2 + OR^2 - 2 \cdot OP \cdot OR \cdot \cos(\angle POR)} \\
 &= \sqrt{2 \left(\frac{1}{\cos(54)} - 1 \right)^2 - 2 \left(\frac{1}{\cos(54)} - 1 \right)^2 \cos(144)} \\
 &\approx 1.334
 \end{aligned}$$